

In class exercises: interaction exercise

This exercise is designed to make sure you understand how to interpret coefficients from interaction models.

You have the following (no noise) data generating process

```
data <- fabricate(
  N = 20,
  X1 = complete_ra(N),
  X2 = block_ra(X1, prob = .6),
  Y = 1 - X1 + 2*X2 - 3*X1*X2,
  X1_demeaned = X1 - mean(X1),
  X2_demeaned = X2 - mean(X2))
```

What is:

1. the effect of X1 when X2 = 0
2. the effect of X2 when X1 = 0
3. the average of X1
4. the average of X2
5. the effect of X1 on the effect of X2
6. the effect of X2 on the effect of X1
7. The average value of Y across all conditions

You run the following regressions:

	Model 1	Model 2
(Intercept)	1.000*** (0.000)	0.800*** (0.000)
X1	-1.000*** (0.000)	
X2	2.000*** (0.000)	
X1:X2	-3.000*** (0.000)	
X1_demeaned		-2.800*** (0.000)
X2_demeaned		0.500*** (0.000)
X1_demeaned:X2_demeaned		-3.000*** (0.000)
R ²	1.000	1.000
Adj. R ²	1.000	1.000
Num. obs.	20	20
RMSE	0.000	0.000

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 1: Statistical models

Which regression model lets you see which of these quantities more easily?

Declare a simple design to calculate the power for quantities 1, 3, and 5. Try to do it from scratch without looking up an existing design.